

Vision-Guided Multi-Objective Reinforcement Learning for UR5 Manipulation in Dynamic Environments

Research Draft

Sections 2–4: Related Work, Contributions, and Expected Results

I. RELATED WORK

Real-world robot manipulation requires a robot to achieve its task while simultaneously considering conflicting objectives such as task success, energy, smoothness, and safety. Existing research has made strong progress in multi-objective reinforcement learning (MORL) and vision-guided manipulation, but these areas have largely developed separately.

The proposed work focuses on the combination of five capabilities:

- 1) **Vision:** RGB-D / point-cloud observations.
- 2) **MORL:** vector rewards and a Pareto-optimal solution set.
- 3) **Manipulation:** contact-rich 6-DoF arm control.
- 4) **Preference conditioning:** runtime modification of behavior using a preference vector λ without retraining.
- 5) **Safety as a constraint:** collision, velocity, and workspace limits formulated as a constrained MDP (CMDP).

Table I provides the high-level comparison.

A. Preference-Conditioned MORL

The MORL literature provides the foundation for learning policies that balance multiple objectives.

Envelope MOQ-learning [6] trains a single network $Q(s, a, \omega)$ over a preference space. Its envelope Bellman operator and hindsight experience replay provide the foundation for single-policy preference conditioning.

PGMORL [5] instead maintains a population of PPO policies trained with different preference vectors. A Gaussian-process prediction model guides the population toward the Pareto front. Hypervolume and sparsity are used to evaluate the resulting solutions.

MO-MPO [8] uses per-objective KL budgets ε_k instead of direct scalar weighting, providing a scale-invariant treatment of multiple objectives.

DOL [11] demonstrates multi-objective learning with image observations through the ImageDeepSea environment. However, the visual environment is a toy grid rather than a camera-based robotic manipulation problem.

MO-Gymnasium [9] provides a standard vector-reward interface for multi-objective environments.

Main limitation: these methods provide the MORL foundation, but do not simultaneously address visual 6-DoF manipulation in dynamic scenes with explicit safety constraints.

B. Vision-Guided Robot Manipulation

Vision-guided manipulation has developed rapidly through imitation learning and visuomotor policy learning.

Diffusion Policy [3] represents the policy as a conditional diffusion process over action sequences. Its reported UR5 setup uses $T_o = 2$, $T_p = 16$, and $T_a = 8$, with a 10 Hz policy and 125 Hz interpolation. The reported velocity limit is < 0.43 m/s and the workspace maintains a position ≥ 1 cm above the table.

3D Diffusion Policy (DP3) [4] uses sparse point-cloud representations instead of RGB images. The method uses 512 or 1024 points and reports a 24.2% relative gain over image baselines on 72 simulation tasks, together with 85% real success with 40 demos.

ACT / ALOHA [12], **Mobile ALOHA** [13], **UMI** [14], and **IRIS** [15] further demonstrate the effectiveness of demonstration-based visuomotor manipulation.

These methods provide the vision and manipulation foundation for the proposed system.

Main limitation: these approaches primarily optimize a single task objective. They do not provide a runtime preference mechanism for changing the trade-off among multiple objectives.

C. Robot-Scale MORL

GCR-PPO [2] provides an important algorithmic foundation for the proposed multi-objective critic.

The method uses a multi-head critic,

$$V_\phi : S \rightarrow \mathbb{R}^K, \quad (1)$$

with component-wise GAE, ratio-preserving advantage normalization, and asymmetric PCGrad.

GCR-PPO reports $\sim 4.55\%$ average return gain over PPO, with up to 9.5% on custom multi-objective suites.

These reported values are used only as an **algorithmic reference** for the scale of improvement observed in the source study. They are not adopted as numerical return targets for the proposed UR5 system.

However, its observations are proprioceptive and the method does not combine a visual encoder, runtime preference conditioning, or explicit CMDP safety.

TABLE I
CAPABILITY COMPARISON BETWEEN REPRESENTATIVE RELATED WORK AND THE PROPOSED APPROACH.

Method	Vision	MORL	Dynamic	Manip.	Preference	Safety Constraint
Envelope MOQ (NeurIPS 2019)	×	✓	×	×	✓	×
PGMORL (ICML 2020)	×	✓	×	×	✓	×
MO-MPO (NeurIPS 2022)	×	✓	×	partial	✓	×
DOL / ImageDeepSea	partial	✓	×	×	✓	×
Diffusion Policy (RSS 2023 / IJRR 2024)	✓	×	✓	✓	×	partial
DP3 (RSS 2024)	✓	×	✓	✓	×	empirical
ACT / ALOHA	✓	×	✓	✓	×	×
GCR-PPO (2025)	×	✓	partial	✓	×	×
PRC-CMORL (Neurocomputing 2023)	×	✓	×	×	✓	✓
Demo-enhanced MORL navigation (IROS 2025)	lidar	✓	✓	×	✓	×
POVNav (2025)	✓	classical	✓	×	partial	✓
This work	✓	✓	✓	✓	✓	✓

D. Safety-Constrained MORL

PRC-CMORL [7] formulates multi-objective robot control as a constrained multi-objective MDP. It provides a foundation for treating safety as a constraint rather than simply another objective.

This distinction is important:

Safety objective $\neq$ Safety constraint.

If safety is included only as a reward term, the policy may trade safety for task performance when λ changes. In contrast, a CMDP explicitly requires safety constraints to remain satisfied within the constraint formulation.

The proposed work therefore uses safety constraints such as

$$d(s, O) \geq D_{\text{safe}}, \quad (2)$$

together with

$$\|\dot{x}\| < 0.43 \text{ m/s}. \quad (3)$$

Safety is therefore handled separately from the preference vector. The preference vector controls the task-objective trade-off among success, energy, and smoothness.

E. Vision and MORL: The Closest Existing Work

Demonstration-enhanced adaptable MO robot navigation [1] is the closest intersection of MORL, preference conditioning, dynamic environments, and sim-to-real.

It uses a preference interpolator

$$I(\lambda) = \lambda^p \quad (4)$$

and adapts the preference at runtime without retraining.

However, the system uses lidar and wheeled robot navigation rather than visual 6-DoF manipulation. Collision is also treated as a sparse objective rather than a CMDP constraint.

POVNav [10] uses Pareto reasoning together with visual information and explicit safety constraints, but its focus is navigation and classical planning rather than a learned preference-conditioned manipulation policy.

F. Research Gap

The literature therefore provides individual building blocks: The resulting research gap is the combination of:

Vision + MORL + 6-DoF Manipulation + Preference Conditioning + CMDP Safety.

The proposed work investigates this combination on a UR5 in static and dynamic environments.

II. CONTRIBUTIONS

The proposed study has three main contributions.

C1. Preference-Conditioned Vision-Guided MORL

The first contribution is a vision-guided MORL framework for UR5 manipulation in which the policy can change its behavior according to a runtime preference vector λ without retraining.

The overall architecture is

$$\begin{array}{c} \text{RGB-D} \rightarrow \text{Point Cloud} \rightarrow \text{DP3} \\ \rightarrow \text{MO Critic} \rightarrow \text{Diffusion Policy} \rightarrow \text{UR5} \end{array} \quad (5)$$

while safety is handled separately through CMDP constraints.

The multi-objective critic uses component-wise GAE, ratio-preserving normalization, and asymmetric PCGrad.

C2. Multi-Objective UR5 Benchmark

The second contribution is a vision-based multi-objective UR5 benchmark containing both static and dynamic environments.

The benchmark considers:

The benchmark is wrapped using the MO-Gymnasium API so that existing MORL baselines can be evaluated using the same environment definition.

TABLE II
WHAT EXISTING RESEARCH PROVIDES AND WHAT REMAINS TO BE COMBINED.

Research Area	What It Provides	Remaining Gap
MORL	Pareto optimization	Limited visual manipulation
Vision manipulation	Visual 6-DoF control	Mostly single-objective
GCR-PPO	Multi-objective critic	No visual preference-conditioned UR5
PRC-CMORL	Safety constraints	No visual manipulation
IROS 2025 MORL navigation	Preference adaptation + dynamic scenes	Navigation rather than manipulation
POVNav	Visual Pareto planning + safety	Classical navigation planning
This work	Combines all five	UR5 dynamic manipulation

TABLE III
PROPOSED UR5 MULTI-OBJECTIVE BENCHMARK STRUCTURE.

Category	Proposed Setting
Robot	UR5
Observation	RGB-D / point cloud
Objectives	Success, energy, smoothness/jerk
Safety	CMDP constraint
Static scene	Static clutter
Dynamic scene 1	Moving obstacle
Dynamic scene 2	Moving target
Dynamic scene 3	Sudden appearance
Preference	Runtime λ

TABLE IV
MAIN EXPERIMENTAL COMPARISONS.

Comparison	Baseline	Purpose
MORL	PGMORL	Pareto performance
MORL	Envelope MOQ	Preference-conditioned MORL
Vision	Single-objective DP3	Visual manipulation reference
Objective	Visual scalarized PPO	Effect of vector reward
Preference	Fixed λ	Effect of runtime preference
Safety	Collision-as-objective	Effect of CMDP constraint

C3. Controlled Evaluation and Sim-to-Real Study

The third contribution is an evaluation protocol designed to separate the effects of:

- 1) multi-objective learning,
- 2) preference conditioning,
- 3) visual representation, and
- 4) safety constraints.

The main comparisons are:

Simulation and real-robot experiments are conducted in parallel so that the sim-to-real gap can be explicitly measured.

III. EXPECTED RESULTS AND COMPARISON

The proposed method is evaluated against published results and against the internal baselines defined for this study. The numbers in the “Expected Result” column are **research targets or reference values, not measured results**. Published results are used to motivate the experimental design, while the proposed targets define the hypotheses to be tested.

A. Overall Expected Comparison

Table V provides the main comparison between reported results from existing methods and the expected results of the proposed Vision-Guided MORL framework.

B. Numerical Targets for Each Research Question

1) *RQ1 — Can the visual representation support multi-objective UR5 control?*: The first experiment verifies whether the DP3 point-cloud representation provides sufficient information for learning the individual objectives before combining them into MORL.

The approximately 92% simulation result is used as the simulation reference for the proposed visual manipulation setup. The real-robot feasibility target is $\geq 70\%$ static success.

The 85% real-robot result reported by DP3 is retained as a **single-objective external reference**. It is not transferred directly to the complete MORL system.

If the individual visual objectives cannot be learned reliably before MORL is introduced, the resulting limitation is likely related to the visual observation or reward formulation rather than the MORL algorithm itself.

2) *RQ2 — Does the multi-head MORL critic improve performance?*: GCR-PPO provides the algorithmic reference for the proposed multi-objective critic.

However, the reported 4.55% average and 9.5% maximum improvements are **not copied as numerical UR5 return targets**. They are used only to establish the order of magnitude of improvement reported for a related multi-objective optimization method.

The scalarized PPO baseline will use the same visual observation setting as the proposed system. Therefore, the comparison does not treat scalarized PPO as a non-visual baseline.

The main expected improvement is therefore evaluated using Pareto-front quality rather than assuming a particular percentage return gain.

The primary metrics are:

- hypervolume,
- Pareto coverage,
- task success,
- energy,
- smoothness, and
- safety-violation rate.

TABLE V
OVERALL EXPECTED COMPARISON BETWEEN PUBLISHED REFERENCES AND THE PROPOSED METHOD.

Method / Reference	Main Result Reported in the Literature	Our Method	Expected Result
DP3 [4]	85% real-robot success with 40 demonstrations	DP3 + MORL + preference conditioning	$\geq 70\%$ real static success target . The 85% DP3 result is retained as a single-objective external reference, not as the target of the complete MORL system.
DP3 [4]	24.2% relative gain of point-cloud representation over image baselines	DP3 point-cloud encoder	Retain the point-cloud representation as the vision backbone; the 24.2% value is used as a representation reference, not as a proposed-system improvement target.
Diffusion Policy [3]	125 Hz UR5 control loop; velocity < 0.43 m/s	Preference-conditioned diffusion policy	Maintain the 125 Hz control loop and the < 0.43 m/s Cartesian velocity constraint.
GCR-PPO [2]	$\sim 4.55\%$ average return improvement; up to 9.5% on selected benchmarks	Multi-head MO critic + asymmetric PCGrad	Use the reported values only as an algorithmic order-of-magnitude reference. The proposed UR5 system is not assigned a 4.55% or 9.5% return target.
PGMORL [5]	Multi-policy Pareto optimization	Single preference-conditioned policy	Hypervolume $\geq$ PGMORL on the same UR5 benchmark wrapper.
Envelope MOQ [6]	Single-model preference-conditioned MORL	Vision + preference-conditioned MORL	Hypervolume $\geq$ Envelope MOQ on the same benchmark wrapper while additionally handling visual manipulation.
PRC-CMORL [7]	Constrained MORL evaluated using Pareto-quality metrics	CMDP-constrained UR5 MORL	Strictly lower safety-violation rate than the collision-as-objective baseline, with static violations approaching 0%.
Proposed Method	Not yet measured	DP3 + MO critic + preference-conditioned diffusion + CMDP	$\geq 70\%$ real static success target; improved Pareto quality; measurable preference-dependent behavior; and reduced safety violations.

TABLE VI
RQ1 VISUAL REPRESENTATION TARGETS.

Metric	Reference / Baseline	Expected Result
Point-cloud size	DP3: 512 or 1024 points	512 points
Demonstrations per task	DP3: 10–20	10–20 demonstrations for warm-start
Simulation success	Proposed visual benchmark	Approximately 92%
Real static success	DP3: 85% with 40 demonstrations	Target: $\geq 70\%$
Point-cloud representation	DP3: 24.2% relative gain	Use point-cloud encoder
Objectives	Proposed benchmark	Success, energy, smoothness
Safety	Conventional reward formulation	Safety treated as a CMDP constraint

The GCR-PPO values of approximately 4.55% and 9.5% are therefore **reference values only**, not guaranteed results for the proposed UR5 system.

3) *RQ3 — Does preference conditioning produce different behaviors without retraining?*: The main purpose of preference conditioning is to allow the same trained policy to produce different behaviors by changing the preference vector λ .

The preference vector is defined as

$$\lambda = [\lambda_{\text{success}}, \lambda_{\text{energy}}, \lambda_{\text{smoothness}}]. \quad (6)$$

Safety is not included in λ . It is handled separately through the CMDP formulation.

For example, the preference settings may emphasize different combinations of the three task objectives:

The important result is not that one preference is universally better. The expected result is that

$$\lambda_i \neq \lambda_j \Rightarrow \text{measurably different robot behavior} \quad (7)$$

without retraining the policy.

Safety is deliberately excluded from this preference table because it is represented by the CMDP constraints rather than by a preference weight.

4) *RQ4 — Does CMDP safety reduce safety violations?*: Safety is treated as a constraint instead of simply adding collision to the reward vector.

The proposed safety constraints are

$$d(s, O) \geq D_{\text{safe}}, \quad (8)$$

$$\|\dot{x}\| < 0.43 \text{ m/s}, \quad (9)$$

TABLE VII
RQ2 MULTI-OBJECTIVE CRITIC COMPARISON.

Method	Average Return	Maximum Gain	Role in Our Study
Scalarized PPO	Baseline	0%	Visual baseline
GCR-PPO	~104.55% of PPO	~109.5%	Literature reference
Proposed MO critic	Not predetermined	Not predetermined	Evaluate Pareto quality

TABLE VIII
RQ3 PREFERENCE-CONDITIONING COMPARISON.

Experiment	Existing Approach	Proposed Approach	Expected Result
Policy training	Separate scalarized policies	One policy	One trained policy
Preference	Fixed scalar weights	Continuous λ	Runtime-adjustable λ
Retraining after preference change	Required in separate-policy setup	Not required	No retraining
Number of tested preferences	Separate policies	$\lambda_1, \dots, \lambda_5$	3–5 settings
Behavior	Fixed trade-off	Preference-dependent	Measurably different behavior
Pareto quality	Baseline	Proposed	HV $\geq$ PGMORL / Envelope target
Real-robot transfer	Individual policies	Preference-conditioned policy	Trade-offs observable on real UR5

TABLE IX
ILLUSTRATIVE PREFERENCE SETTINGS.

Setting	Main Priority	Expected Robot Behavior
λ_1	Task success	Higher success emphasis, potentially higher energy
λ_2	Energy	Lower energy, potentially longer execution
λ_3	Smoothness	Smoother trajectories and motion
λ_4	Balanced	Balanced success–energy–smoothness behavior
λ_5	Custom mixed	User-selected combination of the three objectives

together with workspace limits.

The primary expected result is therefore

$$\boxed{\text{CMDP violation rate} < \text{collision-as-objective violation rate}} \quad (10)$$

with the static validation set expected to approach

$$\boxed{\text{Static safety violation rate} \approx 0\%} \quad (11)$$

The formulation does **not** claim that every safety constraint will always be satisfied. The experimental criterion is a **strictly lower violation rate** than the collision-as-objective baseline, with static violations approaching zero.

A small reduction in task success is acceptable if it results in a substantial reduction in safety violations. The objective is therefore a better constrained Pareto set rather than independently maximizing every metric.

C. Main Quantitative Target Table

The following table summarizes the numerical targets and literature reference values that will be used to determine whether the proposed method achieves its intended improvements.

D. Expected Hypervolume Comparison

Hypervolume is the primary metric for evaluating the quality of the learned Pareto front.

The comparison will use the **same vector-reward UR5 benchmark wrapper** for all principal MORL methods.

The expected relationship is

$$\boxed{HV_{\text{Proposed}} \geq \max(HV_{\text{PGMORL}}, HV_{\text{Envelope}})} \quad (12)$$

when evaluated using the same vector-reward wrapper and benchmark.

The scalarized PPO baseline is explicitly **visual**. It uses the same visual observation setting so that the comparison isolates the effect of multi-objective learning rather than conflating MORL with the use of vision.

The hypervolume comparison is therefore the main criterion for judging whether the proposed MORL formulation provides a useful Pareto advantage.

E. Expected Real-Robot Results

The final experiment transfers the learned policy from simulation to a real UR5.

The DP3 85% real-success result is retained as an external single-objective reference. It should **not** be presented as the expected success rate of the complete proposed MORL system.

The proposed real-robot target is instead

TABLE X
RQ4 SAFETY COMPARISON.

Safety Metric	Collision-as-Objective	Proposed CMDP	Expected Result
Collision handling	Reward penalty	Explicit constraint	Lower violation rate
Minimum obstacle distance	Reward dependent	$d(s, O) \geq D_{\text{safe}}$	Lower violation rate
Cartesian velocity	Not necessarily constrained	< 0.43 m/s	Below 0.43 m/s
Workspace	Reward dependent	Explicit bounds	Reduced workspace violations
Static safety violation rate	Baseline measurement	CMDP	Approaches 0%
Dynamic safety violation rate	Collision objective	CMDP	Lower than collision-as-objective
Task success	Optimized independently	Safety + task objectives	A small reduction is acceptable if safety improves

TABLE XI
MAIN QUANTITATIVE TARGETS AND LITERATURE REFERENCE VALUES.

Performance Area	Published Reference	Reference Value	Proposed Target / Use
Point-cloud representation	DP3 [4]	512 / 1024 points	512 points
Demonstration budget	DP3 [4]	10–20/task	10–20 demonstrations/task
Simulation success	Proposed protocol	Approximately 92%	Approximately 92% simulation reference
Real static success	Proposed protocol	—	$\geq 70\%$ target
DP3 real success	DP3 [4]	85% / 40 demos	Single-objective external reference only
Point-cloud advantage	DP3 [4]	24.2% relative gain	Use point-cloud encoder; not a proposed gain target
Policy/control frequency	Diffusion Policy [3]	125 Hz	125 Hz
Cartesian velocity	Diffusion Policy [3]	< 0.43 m/s	< 0.43 m/s
Preference settings	Proposed protocol	—	3–5 λ settings
Preference adaptation	MORL references	No retraining	No retraining
Preference objectives	Proposed formulation	—	Success, energy, smoothness
Pareto quality	PGMORL / Envelope	Baseline HV	HV $\geq$ PGMORL / Envelope on same wrapper
GCR-PPO improvement	GCR-PPO [2]	$\sim 4.55\%$; up to $\sim 9.5\%$	Algorithmic order-of-magnitude reference only
Static safety violation	Proposed CMDP	—	Approaches 0%
Dynamic safety violation	Collision-as-objective	Baseline measurement	Strictly lower than baseline

TABLE XII
EXPECTED HYPERVOLUME COMPARISON UNDER THE SAME VISUAL VECTOR-REWARD BENCHMARK.

Method	Vision	Number of Policies	Preference	CMDP	Expected Role
Visual Scalarized PPO	✓	Multiple	Fixed	No	Visual scalarized baseline
PGMORL	✓	Multiple	Yes	No	MORL Pareto reference
Envelope MOQ	✓	1	Yes	No	Single-policy preference reference
Proposed Method	✓	1	Continuous λ	Yes	Complete framework

$$\text{Real static success} \geq 70\% \quad (13)$$

while the simulation reference is approximately 92%.

F. What Would Count as Success?

To make the evaluation easy to understand, the proposed method will be considered to meet its principal research objectives if the following conditions are observed.

These criteria convert the research questions into measurable experimental tests rather than qualitative claims.

G. Important Interpretation of the Expected Numbers

The published numbers and the proposed numbers have different meanings.

Therefore, values such as **4.55%**, **9.5%**, **85%**, **24.2%**, **125 Hz**, and **0.43 m/s** must not be presented as measured results of the proposed method before the experiments are performed. In particular:

- The **85% DP3 real-success result** is a single-objective external reference.
- The proposed real static success target is $\geq 70\%$.
- The approximately **92% simulation success** is the simulation reference for the proposed visual manipulation setup.
- The **24.2% DP3 value** supports the point-cloud representation choice and is not a proposed-system gain target.
- The **4.55% and 9.5% GCR-PPO values** provide an algorithmic order-of-magnitude reference only and are not UR5 return targets.
- The primary MORL comparison is based on **hypervolume**

TABLE XIII
EXPECTED REAL-UR5 RESULTS.

Metric	Expected Real-UR5 Result
Control frequency	125 Hz
Policy frequency	10 Hz + interpolation
Maximum Cartesian velocity	< 0.43 m/s
Preference settings	3–5 λ settings
Preference change	No retraining
Preference objectives	Success, energy, smoothness
Real static success	$\geq 70\%$ target
Static safety violations	Approximately 0% target
Dynamic safety violations	Lower than collision-as-objective baseline
Preference-dependent behavior	Measurably different trade-offs
Sim-to-real difference	Success, energy, smoothness, and safety reported separately

TABLE XIV
SUCCESS CRITERIA FOR THE PROPOSED STUDY.

Condition	Target
Visual foundation	Approximately 92% simulation success and $\geq 70\%$ real static success
MORL optimization	Improved Pareto quality relative to visual scalarized PPO
Preference conditioning	3–5 λ values produce measurable behavior changes
No retraining	Preference changes use the same trained policy
Pareto quality	Hypervolume matches or exceeds PGMORL / Envelope on the same benchmark
Safety	Static violation rate approaches 0%
Dynamic safety	CMDP has lower violation rate than collision-as-objective
Real robot	Preference-dependent success–energy–smoothness trade-offs remain observable

TABLE XV
INTERPRETATION OF NUMERICAL VALUES USED IN THE STUDY.

Type	Meaning
Reported	Already measured and reported by the cited paper
Reference	Used to establish the performance level or design choice of an existing method
Target	Experimental threshold or configuration proposed for the present study
Expected	Hypothesized behavior of the proposed system
Measured	To be filled after the actual experiments

and Pareto quality relative to PGMORL, Envelope MOQ, and visual scalarized PPO.

- The preference vector contains only **success, energy, and smoothness**.
- **Safety is handled independently through CMDP constraints.**
- The safety criterion is **lower violation rate**, not a claim that violations can always be completely eliminated.
- Static safety violations are expected to **approach 0%**, while dynamic violations are expected to be **strictly lower than the collision-as-objective baseline**.

After the experiments, the final paper will replace the expected values with the actual measured values and report the corresponding improvement or degradation relative to each baseline.

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